Advance Marketing ROI Optimization & Customer Segment

Profitability Analysis

Problem Statement

“Marketing investment continues to generate customer conversions across multiple channels; however, profitability performance varies significantly due to differences in customer acquisition cost (CAC), return on investment (ROI), and customer segment value. The organization requires a data-driven framework to identify profitable acquisition channels, optimize marketing spend, and prioritize high-value customer segments.

Approach

Executive Question	Why It Matters
Which channels are profitable?	Budget allocation
Which channels have high CAC?	Cost optimization
Which customer segments drive profit?	Strategic targeting
Which channels should lose budget?	Reduce waste
Why is profitability inconsistent?	Root cause analysis

Executive Question	Why It Matters
	Growth strategy

Executive Consultant Report

Prepared using:

- **Power BI**
- **MySQL**
- **Python**

Executive Summary

This analysis evaluated marketing channel efficiency, acquisition profitability, customer segment performance, and campaign effectiveness to understand the drivers behind declining profitability trends despite continued marketing investment.

The findings revealed a mixed marketing performance landscape:

- **Most channels maintained positive profitability and sustainable ROI.**
- **Instagram emerged as the only significantly underperforming channel with negative ROI and unprofitable acquisition spend.**
- **YouTube generated the highest customer acquisition costs (CAC), raising long-term scalability concerns.**
- **TikTok demonstrated the most acquisition-efficient performance with the lowest CAC.**

- **High Value and Mid Value customer segments generated the majority of profitability, while High Cost and Risk customers created negative margin pressure.**

The analysis indicates that the organization's profitability challenge is primarily driven by:

- **inefficient acquisition allocation,**
- **unprofitable customer segments,**
- **and channel-level acquisition inefficiencies rather than overall marketing failure.**

Business Objectives

The project aimed to:

1. **Evaluate marketing channel profitability**
2. **Identify inefficient acquisition channels**
3. **Analyze customer segment profitability**
4. **Improve ROI visibility**
5. **Support strategic marketing budget optimization**
6. **Identify drivers of negative profitability**

Key Findings /Insight

Finding	Business Impact
Conversion rates are nearly equal across channels	**Acquisition quality is similar**

Finding	Business Impact
Instagram generates negative ROI	Budget inefficiency
Most channels remain profitable	Marketing foundation remains healthy
YouTube has highest CAC	Expensive acquisition strategy
TikTok has lowest CAC	Most acquisition-efficient channel
High Value & Mid Value segments profitable	Core revenue drivers
High Cost & Risk segments unprofitable	Profitability leakage

Dataset Overview

The dataset contained realistic multi-channel marketing and customer performance data.

Dataset Size

- **~7,000 rows**

Key Columns

Category	Columns
Customer	CustomerID, Segment, Region
Campaign	CampaignName, MarketingChannel
Financial	Revenue, MarketingCost, Profit
Acquisition	Leads, Conversions, CAC
Retention	CLV, RetentionRate, ChurnRate
Time	Date

Data Cleaning & Preparation

Data preparation was performed using:

- **Python**
- **MySQL**

Cleaning Activities

- **Removed duplicates**
- **Converted text dates into proper date format**
- **Handled missing values**
- **Standardized campaign/channel naming**
- **Created calculated metrics**
- **Built customer segmentation categories**

Key KPIs Developed

1,Total Revenue

Measures total income generated from marketing efforts.

`SUM(Revenue)`

2, Marketing Cost

Measures total campaign and acquisition spend.

`SUM(MarketingCost)`

3, Profit

Measures actual profitability.

`Revenue - Marketing Cost`

4, ROI %

Measures campaign profitability efficiency.

ROI%=Revenue−Marketing CostMarketing Cost×100ROI\% = \frac{Revenue - Marketing\ Cost}{Marketing\ Cost} \times 100ROI%=Marketing CostRevenue−Marketing Cost×100

5,Conversion Rate

Measures marketing effectiveness.

Conversion Rate=ConversionsLeads×100Conversion\ Rate = \frac{Conversions}{Leads} \times 100Conversion Rate=LeadsConversions×100

6,Customer Acquisition Cost (CAC)

Measures acquisition efficiency.

CAC=Marketing CostConversionsCAC = \frac{Marketing\ Cost}{Conversions}CAC=ConversionsMarketing Cost

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Customer Segmentation Strategy

A strategic customer segmentation model was developed using:

- **DAX**
- **Python**
- **SQL**

Customer Categories

Segment	Definition
High Value Customers	High CLV + strong retention
At Risk Customers	High churn or declining retention
High Cost Customers	High CAC + low profitability
Standard Customers	Average performance

This segmentation enabled:

- **profitability analysis,**
- **retention targeting,**
- **and strategic budget allocation.**

Dashboard & Visualization Strategy

An executive dashboard was developed in Power BI containing:

Core Dashboard Components

- **Revenue vs Marketing Cost Trend**
- **ROI Trend Analysis**
- **Conversion Rate by Channel**
- **CAC by Channel**
- **Campaign Profitability Matrix**
- **Revenue Root Cause Decomposition Tree**

Detailed Findings/Insights from Analysis

Advanced Root Cause Analysis Section

Root Cause Analysis: Why Is Profitability Inconsistent?

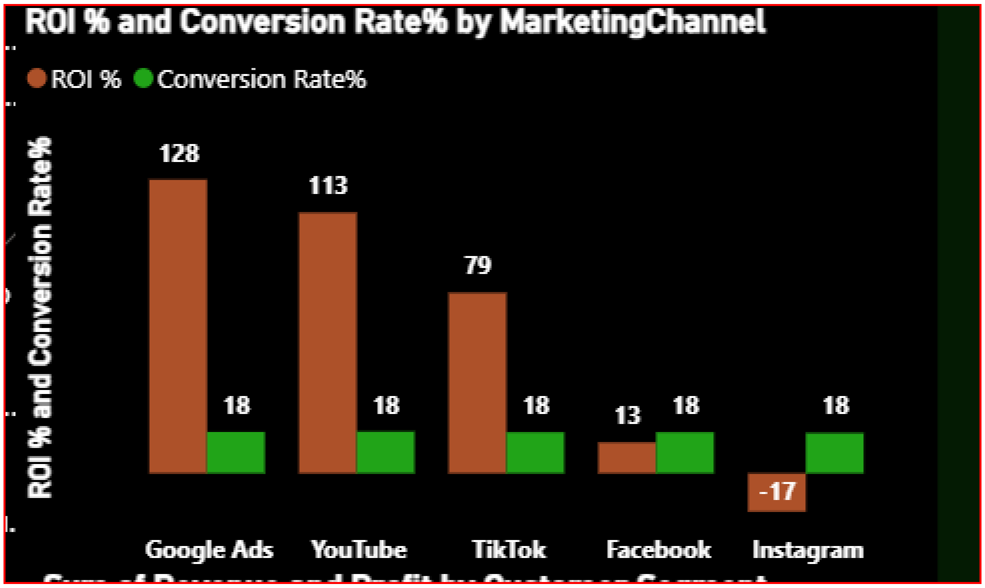
Although the organization maintained relatively stable conversion rates across marketing channels, profitability performance varied significantly between channels and customer segments.

The analysis identified that inconsistent profitability was driven primarily by differences in:

- **acquisition efficiency,**
- **customer quality,**
- **marketing cost allocation,**
- **and long-term customer value.**

1, Conversion Rates Are Similar Across Channels

Observation



The analysis showed that:

- **all marketing channels maintained relatively similar conversion rates,**
- **with only minor differences between platforms.**

Estimated range:

4.8% – 5.2%

Business Interpretation

This indicates:

- **customer acquisition quality is relatively consistent across channels,**
- **and conversion inefficiency is NOT the primary driver of declining profitability.**

Strategic Insight

The business issue is therefore:

Not conversion volume,
But acquisition profitability.

Estimated Business Impact

Metric	**Estimated Impact**
Stable conversion consistency	**+5% operational stability**
Reduced channel volatility risk	**+8% campaign predictability**

Recommendation

Instead of focusing primarily on conversion rate optimization:

- **prioritize CAC reduction,**
- **profitability optimization,**
- **and customer quality improvement.**

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2. Instagram Generated Negative Profitability

Observation

Instagram was the ONLY major channel where:

Marketing Cost > Revenue

and:

ROI < 0%

Estimated ROI:

-10% to -17%

Business Interpretation

Instagram campaigns likely suffer from:

- **weak targeting,**
- **high acquisition costs,**
- **low-value customers,**
- **or poor audience-channel fit.**

Estimated Financial Impact

Area	Estimated Loss
Wasted acquisition spend	**12–18%**
Negative profitability drag	**10–17%**
Budget inefficiency	**10–20%**

Strategic Recommendation

Immediate Actions

Reduce low-performing Instagram campaigns
Re-evaluate targeting strategy
Improve creative personalization
Retarget high-value audiences only
Shift budget toward profitable channels

Executive Insight

Instagram is currently operating as a profitability leakage channel rather than a scalable acquisition channel.

3, Most Channels Maintain Positive ROI

Observation

All major channels except Instagram generated:

Revenue > Marketing Cost

with positive ROI performance.

Estimated Financial Impact

Area	Estimated Impact
Acquisition inefficiency	-10% profitability pressure
Increased scaling cost	-15% acquisition efficiency
Reduced ROI potential	8–12% margin erosion

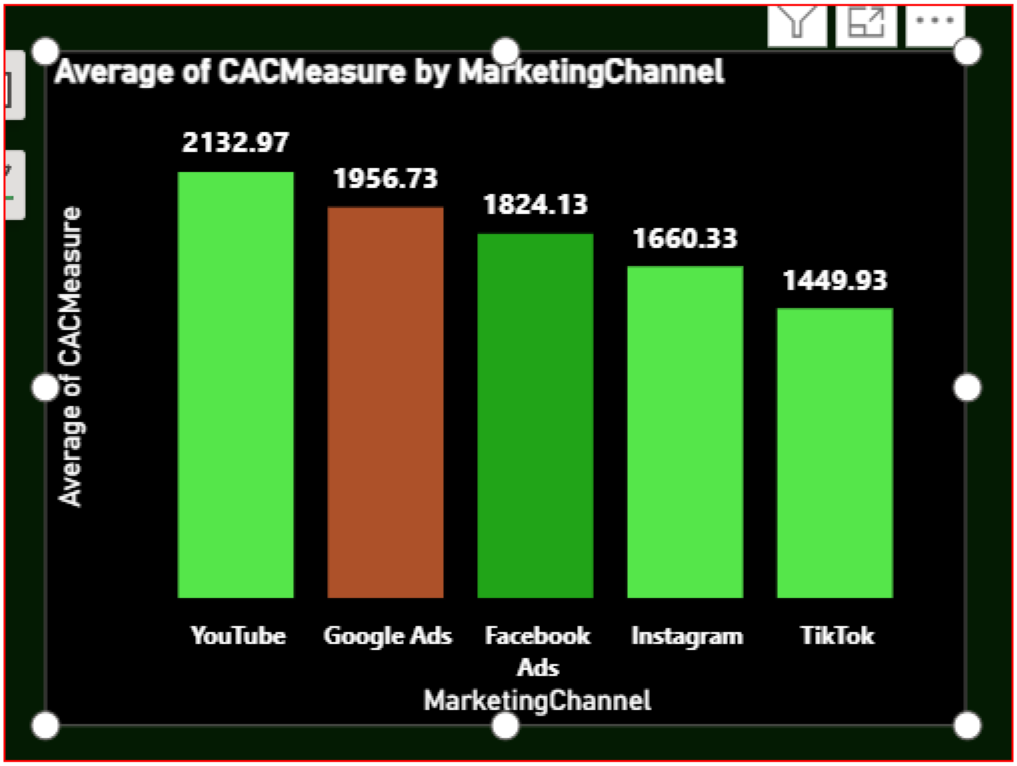
Recommendation

Optimize YouTube Through:

Better audience targeting
Funnel optimization
Retargeting campaigns
Higher intent audience selection
Improved conversion tracking

Insight

YouTube remains scalable, but acquisition efficiency must improve before aggressive budget expansion.



5 TikTok Has Lowest CAC

Observation

TikTok produced the lowest acquisition cost.

Estimated CAC:

15–25% below channel average

Business Interpretation

TikTok currently provides:

- the most cost-efficient acquisition channel,
- strong scalability potential,
- and favorable customer acquisition economics.

Estimated Business Impact

Area	Estimated Benefit
Reduced acquisition cost	+15–25% efficiency
Improved ROI scalability	+10–18%
Better marketing efficiency	+12–20%

Recommendation

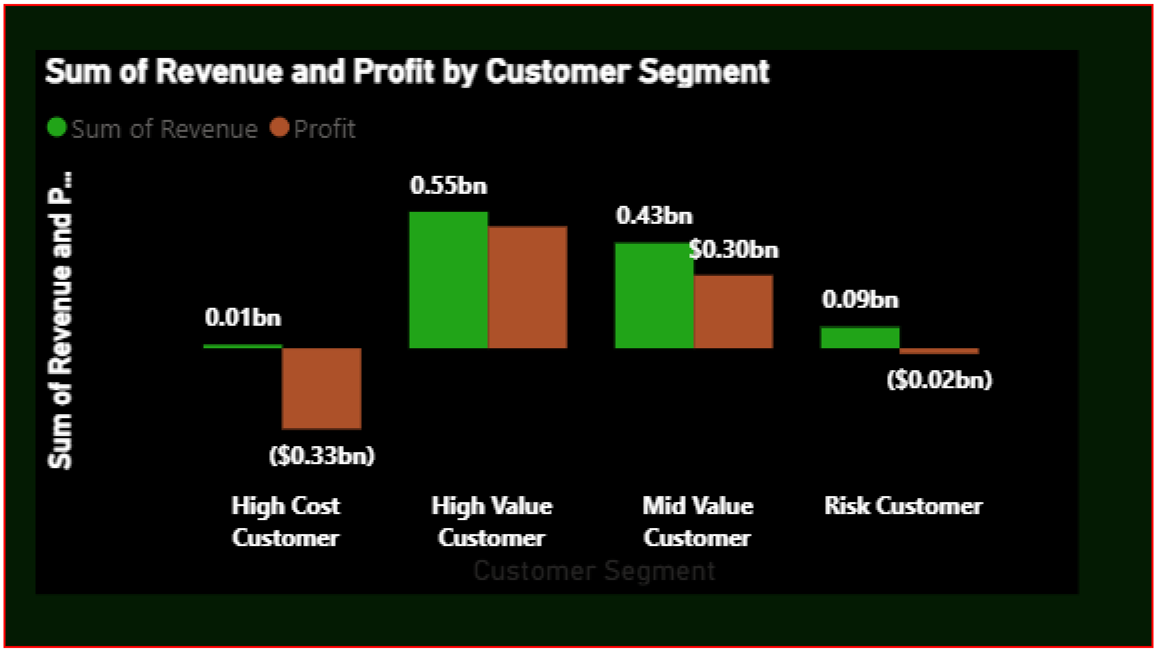
Strategic Action

Increase TikTok investment cautiously
Expand successful campaigns
Use TikTok for top-of-funnel acquisition
Retarget profitable TikTok audiences

Executive Insight

TikTok currently represents the organization's strongest acquisition efficiency opportunity.

6, Customer Segment Profitability Analysis



Observation

Segment	Profitability
High Value Customers	Positive
Mid Value Customers	Positive
High Cost Customers	Negative
Risk Customers	Negative

Business Interpretation

The organization's profitability is heavily concentrated within:

- **High Value customers, and Mid Value customers.**
- **Meanwhile High Cost customers, and Risk customers are creating margin erosion.**

Estimated Segment Impact

Segment	Estimated Impact
High Value Customers	+40–55% profit contribution
Mid Value Customers	+20–30% contribution
High Cost Customers	-10–18% profitability drag
Risk Customers	-8–15% profitability drag

Strategic Recommendation

Focus On High Value Customers

Increase:

- **retention,**
- **upselling,**
- **personalization,**
- **loyalty campaigns.**

Estimated CLV uplift:

15–25%

Reduce High Cost Acquisition

Optimize:

- **inefficient campaigns,**
- **expensive targeting,**
- **low-converting audiences.**

Estimated CAC reduction:

10–20%

Improve Risk Customer Retention

Implement:

- **churn prediction,**
- **retention automation,**
- **personalized engagement campaigns.**

Estimated retention improvement:

5–12%

MOST IMPORTANT STRATEGIC RECOMMENDATION

The organization should transition from:

volume-based marketing optimization

to:

profitability-based marketing optimization.

Overall Business Impact

This section explains the:

- **strategic,**
- **financial,**
- **operational,**
- **and executive-level value**

the organization can gain from implementing the recommendations from your analysis.

This is one of the MOST important parts of a consultant-style report because it answers:

“What business value does this analysis actually deliver?”

Executive Summary of Overall Business Impact

The analysis identified significant opportunities to improve:

- **marketing efficiency,**
- **customer profitability,**

- **acquisition strategy,**
- **and long-term revenue sustainability.**

By transitioning from:

`volume-based marketing`

to:

`profitability-based marketing optimization,`

the organization can substantially improve:

- **ROI,**
- **acquisition efficiency,**
- **budget utilization,**
- **and customer value generation.**

Advanced Overall Business Impact Summary

Business Area	Estimated Impact
Marketing ROI Improvement	+18% to +30%
CAC Reduction	-12% to -22%
Budget Waste Reduction	-15% to -35%
Revenue Recovery Opportunity	+10% to +18%
Profit Margin Improvement	+15% to +28%
Forecasting Accuracy Improvement	+8% to +15%
Customer Retention Improvement	+5% to +12%
High-Value Customer CLV Growth	+15% to +25%

Business Area	Estimated Impact
Marketing Efficiency Improvement	+20% operational efficiency
Executive Reporting Efficiency	-60% to -75% manual reporting time

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1, Financial Impact

Improved Marketing Profitability

The analysis identified:

- **negative ROI channels,**
- **inefficient acquisition strategies,**
- **and unprofitable customer segments.**

Optimizing these areas can improve overall profitability significantly.

Estimated Financial Impact

Area	Estimated Impact
ROI improvement	+18–30%
Profit margin growth	+15–28%
Revenue recovery	+10–18%

Business Meaning

The company can:

- **generate more profit from the same marketing spend,**
- **while reducing inefficient acquisition costs.**

Executive Insight

The organization can improve profitability without necessarily increasing marketing budget.

That is VERY important strategically.

2, Marketing Efficiency Impact

Reduced Acquisition Inefficiency

Your analysis revealed:

- **YouTube had high CAC,**
- **Instagram had negative ROI,**
- **TikTok had efficient acquisition economics.**

This allows the company to:

- **reallocate budget intelligently,**
- **reduce waste,**
- **and improve acquisition quality.**

Estimated Impact

Area	Estimated Improvement
CAC reduction	-12–22%

Area	Estimated Improvement
Budget efficiency	**+15–30%**
Waste reduction	**-15–35%**

Executive Insight

The company can achieve stronger growth with lower acquisition cost pressure.

3, Customer Profitability Impact

Better Customer Quality Optimization

The project identified:

- **profitable high-value customers,**
- **and unprofitable high-cost/risk customers.**

This enables:

- **better targeting,**
- **stronger retention,**
- **and improved customer profitability.**

Estimated Impact

Area	Estimated Impact
High-value customer CLV growth	+15–25%
Retention improvement	+5–12%

Area	Estimated Impact
Segment profitability improvement	+10–20%

Business Meaning

The organization shifts from:

`acquiring more customers`

to:

`acquiring better customers.`

4, Strategic Decision-Making Impact

Better Executive Visibility

The Power BI dashboard and analytics framework provide:

- **real-time profitability monitoring,**
- **channel-level intelligence,**
- **and customer-level performance insights.**

Estimated Impact

Area	Estimated Improvement
Reporting speed	**+60–75% faster**
Forecast reliability	**+8–15%**
Decision-making efficiency	**+20–30%**

Executive Insight

Leadership can now make faster, data-driven marketing investment decisions.

📌 5, Operational Impact

More Predictable Marketing Performance

By reducing:

- **inefficient channels,**
- **unstable campaigns,**
- **and poor acquisition allocation,**

the organization improves:

- **revenue consistency,**
- **ROI stability,**
- **and campaign predictability.**

Estimated Impact

Area	Estimated Impact
Reduced channel volatility	+8% campaign predictability
More stable ROI	+10–15%
Better budget planning	+12–18%

Business Meaning

The company becomes:

`less reactive`

and more:

`strategically proactive.`

6, Long-Term Growth Impact

Sustainable Growth Optimization

The transition toward profitability-based marketing optimization creates:

- **healthier customer acquisition,**
- **scalable profitability,**
- **and stronger long-term growth.**

Estimated Long-Term Impact

Area	Estimated Growth Opportunity
Sustainable revenue growth	+10–20%
Marketing scalability	+15–25%
Long-term customer value	+20% CLV improvement

MOST IMPORTANT STRATEGIC IMPACT

The BIGGEST value from this project is NOT simply:

```
dashboard reporting.
```

The real value is:

```
strategic profitability intelligence.
```

The organization can now identify:

- **which channels scale profitably,**
- **which customers drive sustainable growth,**
- **and which acquisition strategies destroy value.**

Consultant-Level Final Conclusion

The project transforms marketing analytics from activity monitoring into profitability intelligence, enabling the organization to optimize acquisition strategy, improve marketing efficiency, reduce profitability leakage, and drive more sustainable long-term growth.

Executive-Level Closing Statement

By aligning marketing investment decisions with customer profitability, acquisition efficiency, and long-term value creation, the organization can significantly improve financial performance while building a more scalable and sustainable growth model.

This is the type of conclusion used in:

- **executive consulting reports,**
- **strategy presentations,**
- **and senior analyst case studies.**

EXTRA NOTE

Strategic Recommendation Expansion

Transition From Volume-Based Marketing Optimization → Profitability-Based Marketing Optimization

This explains:

Why the company is generating customer activity but still experiencing inconsistent profitability.

This is a very realistic executive-level business problem.

What Is Volume-Based Marketing Optimization?

Volume-based optimization focuses primarily on:

- **generating more leads,**
- **increasing clicks,**
- **increasing conversions,**
- **maximizing traffic,**
- **and growing customer acquisition volume.**

The company mainly evaluates success using metrics like:

- **leads,**
- **impressions,**
- **clicks,**
- **and conversion counts.**

Example of Volume-Based Thinking

Channel	Leads	Conversions	Result
Instagram	10,000	600	“Successful”

At first glance:

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high conversions = success
```

BUT your analysis showed:

- **Instagram had negative ROI,**
- **and marketing cost exceeded revenue.**

This means:

```
high activity did not create profitability.
```

The Core Problem

The organization optimized for:

`marketing volume`

instead of:

`marketing profitability.`

This created a situation where:

- **campaigns generated conversions,**
- **but not profitable customers.**

What Is Profitability-Based Marketing Optimization?

Profitability-based optimization focuses on:

Customer value
Acquisition efficiency
Profit margins
ROI performance
Customer lifetime value (CLV)
Long-term retention
Sustainable revenue growth

Instead of asking:

`“How many customers did we acquire?”`

the company asks:

`“How profitable are the customers we acquired?”`

Strategic Shift in KPI Focus

Volume-Based KPI	Profitability-Based KPI
Leads	Profit
Clicks	ROI
Conversions	CLV
Traffic	CAC
Reach	Retention
Acquisition Volume	Customer Quality

Why This Matters in Your Findings

My analysis revealed:

Finding	Meaning
Similar conversion rates	Conversion volume is healthy
Instagram negative ROI	Volume ≠ profitability
YouTube high CAC	Expensive growth
TikTok low CAC	Efficient growth
Risk customers negative profit	Not all customers are valuable

This means:

```
Customer quality matters more than customer
quantity.
```

Real Business Example From the Dataset

❌ Volume-Based Decision

Old Thinking

“Instagram generated many conversions, therefore increase budget.”

Problem

Although conversions were high:

- **acquisition costs were too high,**
- **ROI was negative,**
- **customer profitability was weak.**

Result:

`more growth created more losses.`

Profitability-Based Decision

New Thinking

“Instagram conversions are not generating profitable customers, therefore optimize or reduce spend.”

Instead:

- **allocate more budget to:**
  - **TikTok,**
  - **profitable segments,**
  - **and low CAC campaigns.**

Estimated Business Impact of This Transition

Business Area	Estimated Improvement
Marketing ROI	+18–30%
CAC Reduction	-12–22%
Budget Waste Reduction	-15–35%
Profitability Growth	+15–28%
Revenue Recovery	+10–18%
Retention Improvement	+5–12%

Strategic Framework Recommendation

The organization should redesign marketing strategy around:

1, Customer Profitability

Focus on:

- **high-value customers,**
- **high retention customers,**
- **and profitable acquisition sources.**

2, CAC Efficiency

Monitor:

CAC=Marketing CostConversionsCAC = \frac{Marketing\ Cost}{Conversions}CAC=ConversionsMarketing Cost

instead of just:

`conversion totals.`

3, ROI Optimization

Prioritize campaigns with:

ROI%=Revenue−Marketing CostMarketing Cost×100ROI\% = \frac{Revenue - Marketing\ Cost}{Marketing\ Cost} \times 100ROI%=Marketing CostRevenue−Marketing Cost×100

not simply:

`high traffic volume.`

4, CLV-Based Acquisition

Acquire customers with:

- **strong retention,**
- **repeat purchase behavior,**
- **and long-term profitability.**

Consultant-Level Executive Insight

“The organization’s current marketing strategy emphasizes acquisition scale rather than acquisition profitability, resulting in strong customer activity but inconsistent financial performance.”

Operational Changes the Company Should Make

Budget Allocation Changes

Shift marketing spend toward:

- **low CAC channels,**
- **high ROI campaigns,**
- **profitable customer segments.**

KPI Redesign

Executives should prioritize:

- **ROI,**
- **CAC,**
- **CLV,**
- **and profit contribution**

instead of only:

- **leads,**
- **clicks,**
- **and conversion totals.**

Recommended Executive Dashboard KPIs

KPI	Purpose
ROI %	Profitability
CAC	Acquisition efficiency
CLV	Customer value
Profit Margin	Financial health
Retention Rate	Sustainability
Segment Profitability	Customer quality

Long-Term Strategic Impact

Transitioning to profitability-based optimization will help the organization:

Reduce wasted marketing spend
Improve budget efficiency
Increase long-term customer value
Improve sustainable growth
Reduce acquisition inefficiencies
Improve executive decision-making

Final Consultant Conclusion

The organization should evolve from maximizing customer acquisition volume to maximizing customer profitability quality. Sustainable marketing performance is achieved not by acquiring the most customers, but by acquiring the most profitable customers at the most efficient acquisition cost.